Generating functions

NGUYEN Hoang Thach
nhthach@math.ac.vn

Outline

1. Generating functions and formal power series
2. Counting with generating functions
3. Generating functions and recurrence relation

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Generating function of a sequence

Definition

The generating function *for the sequence* $\{a_n\}_{n\geq 0}$ *is the formal power series:*

$$G_a(x) = a_0 + a_1 x + a_2 x^2 + \ldots + a_n x^n + \cdots = \sum_{n=0}^{\infty} a_n x^n .$$

Remark:

- The power series is *formal*, meaning that we generally ignore questions such as convergence of the series.
- A finite sequence $a_0, a_1, \ldots, a_N$ can be extended into an infinite sequence by considering that $a_k = 0$ for all $k > N$.

Examples of generating functions

- $a_n = 1$ for all n:

$$G_a(x) = 1 + x + x^2 + \cdots = \sum_{n=0}^{\infty} x^n \,.$$

- $b_n = n$ for all n:

$$G_b(x) = x + 2x^2 + 3x^3 + \cdots = \sum_{n=1}^{\infty} nx^n \,.$$

- $\{c_n\}$ is the finite sequence $1, 5, 10, 10, 5, 1$:

$$G_c(x) = 1 + 5x + 10x^2 + 10x^3 + 5x^4 + x^5 = (1 + x)^5 \,.$$

Operations of formal power series

Theorem

Let $F(x) = \sum_{n=0}^{\infty} a_n x^n$, $G(x) = \sum_{n=0}^{\infty} b_n x^n$ *and* $\lambda \in \mathbb{R}$. *Then:*

$$F(x) + G(x) = \sum_{n=0}^{\infty} (a_n + b_n) x^n \,, \tag{1}$$

$$\lambda F(x) = \sum_{n=0}^{\infty} (\lambda a_n) x^n \,, \tag{2}$$

$$F(x)G(x) = \sum_{n=0}^{\infty} \left(\sum_{i=0}^{n} a_i b_{n-i} \right) x^n \,, \tag{3}$$

$$F'(x) = \sum_{n=1}^{\infty} n a_n x^{n-1} = \sum_{n=0}^{\infty} (n+1) a_{n+1} x^n \,. \tag{4}$$

Example

The *Maclaurin series* is the generating function of the sequence $a_n = 1$ for all n:

$$G(x) = \sum_{n=0}^{\infty} x^n = \frac{1}{1-x}\,.$$

Taking the derivative:

$$\sum_{n=0}^{\infty}(n+1)x^n = G'(x) = \left(\frac{1}{1-x}\right)' = \frac{1}{(1-x)^2}\,.$$

Alternatively:

$$\begin{aligned}\frac{1}{(1-x)^2} = G(x)G(x) &= \left(\sum_{n=0}^{\infty} x^n\right)\left(\sum_{n=0}^{\infty} x^n\right)\\ &= \sum_{n=0}^{\infty}\left(\sum_{k=0}^{n} x^k x^{n-k}\right)\\ &= \sum_{n=0}^{\infty}(n+1)x^n\,.\end{aligned}$$

Extended binomial coefficients

Definition

Let $\alpha \in \mathbb{R}$ *and* $k \in \mathbb{N}$. *The* extended binomial coefficient $\binom{\alpha}{k}$ *is defined by:*

$$\binom{\alpha}{k} = \begin{cases} \alpha(\alpha-1)\cdots(\alpha-k+1)/k! \text{ if } k > 0\,, \\ 1 \text{ if } k = 0\,. \end{cases} \tag{5}$$

Examples:

- For positive integers n and k:

$$\binom{-n}{k} = (-1)^k \binom{n+k-1}{k}\,.$$

- For k positive integer:

$$\binom{1/2}{k} = \frac{(-1)^{k-1}}{4^{k-1}2k}\binom{2k-2}{k-1}\,.$$

The extended binomial formula

Theorem

For all $\alpha \in \mathbb{R}$:

$$(1+x)^{\alpha} = \sum_{n=0}^{\infty} \binom{\alpha}{n} x^n \,. \tag{6}$$

Examples:

- $(1-x)^{-2}$:

$$(1-x)^{-2} = \sum_{n=0}^{\infty} \binom{-2}{n} (-x)^n = \sum_{n=0}^{\infty} (n+1) x^n \,.$$

- $\sqrt{1+x}$:

$$\sqrt{1+x} = (1+x)^{1/2} = \sum_{n=0}^{\infty} \binom{1/2}{n} x^n = 1 + \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{4^{n-1}(2n)} \binom{2n-2}{n-1} x^n$$

Some useful power series

$$(1+x)^k = \sum_{n=0}^{k} \binom{k}{n} x^n$$

$$\frac{1}{1-ax} = \sum_{n=0}^{\infty} a^n x^n$$

$$\frac{1}{1-x^k} = \sum_{n=0}^{\infty} x^{kn}$$

$$\frac{1}{(1-x)^k} = \sum_{n=0}^{\infty} \binom{n+k-1}{n} x^n$$

$$e^x = \sum_{n=0}^{\infty} \frac{1}{n!} x^n$$

$$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n$$

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Counting with generating functions

Find the number of ways to make \$7 from \$1, \$2 and \$5 bills if order does not matter.

The values (in dollars) that can be made with \$1 bills are 0, 1, 2, ... The corresponding generating function is $1 + x + x^2 + \cdots$

The values that can be made with \$2 bills are 0, 2, 4, ... The corresponding generating function is $1 + x^2 + x^4 + \cdots$
The values that can be made with \$5 bills are 0, 5, 10, ... The corresponding generating function is $1 + x^5 + x^{10} + \cdots$

The generating function for the number of ways to make n dollars is the coefficient of x^n in the generating function

$$(1 + x + x^2 + \cdots)(1 + x^2 + x^4 + \cdots)(1 + x^5 + x^{10} + \cdots).$$

Ans: $a_7 = 6$.

Counting with generating functions

Find the number of integer solutions to $x_1 + x_2 + x_3 = 10$ *such that* $2 \leq x_1 \leq 4$, $3 \leq x_2 \leq 5$ *and* x_3 *is divisible by 3.*

The generating function for x_1 is $x^2 + x^3 + x^4$.
The generating function for x_2 is $x^3 + x^4 + x^5$.
The generating function for x_3 is $1 + x^3 + x^6 + x^9 + \cdots$.
Let $G(x) = (x^2 + x^3 + x^4)(x^3 + x^4 + x^5)(1 + x^3 + x^6 + x^9 + \cdots)$.
The number of solutions is the coefficient of x^{10} in the expasion of G.

Ans: 10.

Counting with generating functions

Find the number of ways to put n identical balls into k labeled boxes so that none of the boxes is empty.

The generating function for the number of balls in each box is $x + x^2 + x^3 + \cdots$

Let $G(x) = (x + x^2 + x^3 + \cdots)^k$.

The number of ways to put n identical balls into k labeled boxes so that none of the boxes is empty is the coefficient of x^n and is $\binom{n-1}{n-k}$.

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The Fibonacci sequence

$$\begin{cases} f_n = f_{n-1} + f_{n-2} \text{ for } n \geq 2\,, \\ f_0 = 0, f_1 = 1\,. \end{cases}$$

Let $F(x) = \sum_{n=0}^{\infty} f_n x^n$.

Then:

$$F(x) = x + x\left(F(x) - f_0\right) + x^2 F(x)\,.$$

Solve for F:

$$F(x) = \frac{x}{1 - x - x^2}\,.$$

The Fibonacci sequence

Let $r_1 = (-1+\sqrt{5})/2$ and $r_2 = (-1-\sqrt{5})/2$. We need to find A, B such that:

$$\frac{x}{1-x-x^2} = \frac{A}{x-r_1} + \frac{B}{x-r_2},$$

Solve for A and B:

$$A = \frac{-r_1}{r_1 - r_2}, \quad B = \frac{r_2}{r_1 - r_2}.$$

Hence:

$$F(x) = \frac{1}{r_1 - r_2}\left(\frac{1}{1-\frac{x}{r_1}} - \frac{1}{1-\frac{x}{r_2}}\right) = \frac{1}{r_1 - r_2}\sum_{n=0}^{\infty}\left(\left(\frac{1}{r_1}\right)^n - \left(\frac{1}{r_2}\right)^n\right)x^n$$

and

$$f_n = \frac{1}{r_1 - r_2}\left(\left(\frac{1}{r_1}\right)^n - \left(\frac{1}{r_2}\right)^n\right) = \frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n - \frac{1}{\sqrt{5}}\left(\frac{1-\sqrt{5}}{2}\right)^n$$

The sum of the first integers

Let $s_n = 1 + 2 + \cdots + n$. We have:

$$\begin{cases} s_n = s_{n-1} + n \text{ for } n \geq 1\,, \\ s_0 = 0\,. \end{cases}$$

Let $S(x) = \sum_{n=0}^{\infty} s_n x^n$.

Then:

$$S(x) = xS(x) + \frac{x}{(1-x)^2}\,,$$

and:

$$S(x) = \frac{x}{(1-x)^3}\,,$$

Derivating $(1-x)^{-2}$ or using the extended binomial formula, we get:

$$S(x) = \sum_{n=0}^{\infty} \frac{n(n+1)}{2} x^n\,.$$